

HACKUP TECHNOLOGY PVT. LTD.

Ethical Hacking & Cyber Security

Capstone Project Report

SOC and Digital Forensics Investigation System

AI-Powered SOC & Digital Forensics Internship

Program: Ethical Hacking & Cyber Security

Duration: 30 Days | Week 1 - Week 5

Year: 2026

By
Vijitha s (24233)
B.sc.,cs(SSS)

Capstone Project

Title: SOC and Digital Forensics Investigation

System Students are required to:

- Analyze logs using a SIEM tool
- Perform disk analysis using forensic tools
- Analyze memory data
- Generate an AI-assisted incident report

Deliverables:

- Investigation report 
- Screenshots and evidence
- GitHub repository submission

Tools Used (Primarily Open-Source)Lab Environment

- Kali Linux
- VirtualBox

SIEM and Monitoring

- Wazuh
- ELK Stack (Elasticsearch, Logstash, Kibana)

Log Analysis

- Linux utilities (grep, cat, tail)

Network Analysis

- Wireshark

Digital Forensics

- Autopsy
- Volatility Framework

Supporting Tools (Optional)

- FTK Imager
- ExifTool

AI Support

- ChatGPT (for analysis and reporting assistance)

Introduction

This report documents the Capstone Project completed as part of the AI-Powered SOC & Digital Forensics internship at Hackup Technology Pvt. Ltd. The project integrates all skills acquired during the 5-week program including SIEM configuration, log analysis, network packet analysis, digital forensics, and AI-assisted reporting.

The objective was to simulate a real-world SOC environment, detect suspicious activities, perform disk and memory forensics, and generate a comprehensive incident investigation report.

Tools Used

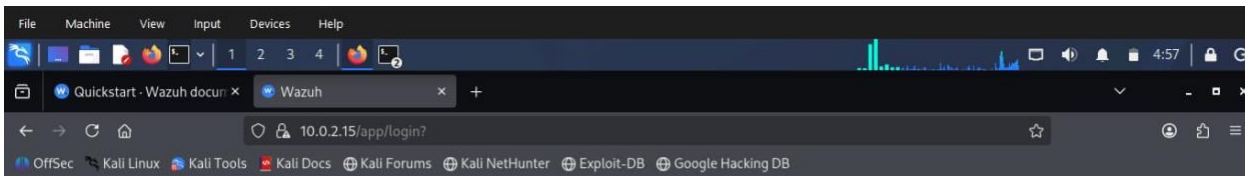
- **Wazuh SIEM** – Used to monitor security events and detect alerts.
- **Kibana / ELK Stack** – Used to view and search logs.
- **Wireshark** – Used to capture and analyze network traffic.
- **Autopsy Forensic Browser** – Used for disk forensic investigation.
- **Volatility Framework** – Used for memory analysis.
- **Kali Linux (VirtualBox)** – Used as the lab environment.
- **Linux Commands (grep, cat, tail, ping, find)** – Used for log analysis and system investigation.

SIEM Setup & Monitoring (Wazuh)

Wazuh was used as the main SIEM tool for monitoring security events. The Kali Linux system was added as an agent, and security logs were collected and analyzed through the Wazuh dashboard.

Wazuh Dashboard Loading

Initial setup of the Wazuh SIEM dashboard, accessible at IP 10.0.2.15.



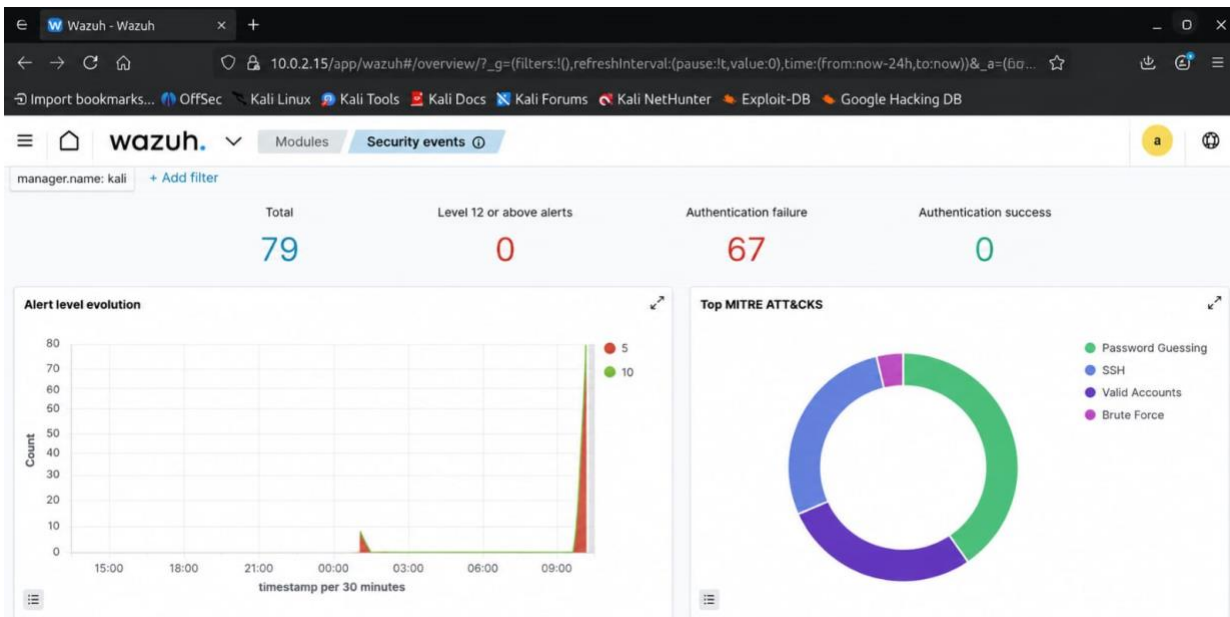
wazuh.

Loading ...

Wazuh SIEM Loading Screen

Security Events Overview - Initial Monitoring

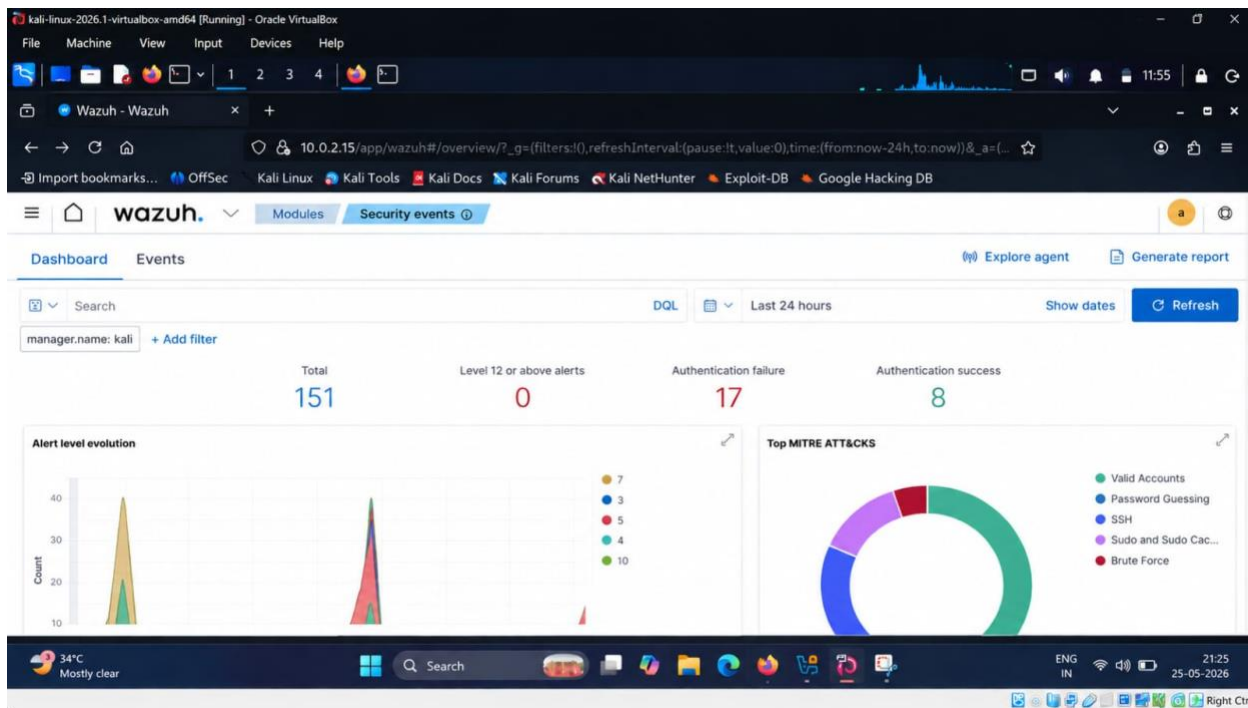
The Wazuh dashboard showed the security events collected from the Kali Linux system. It displayed login failures and other security alerts. The dashboard helped monitor system activity and identify possible security threats.



Wazuh Security Events - 79 Total Alerts, 67 Auth Failures

Security Events Dashboard - Extended Monitoring

The Wazuh dashboard recorded security events, login failures, and successful logins. It helped monitor system activity and identify suspicious behavior.



Wazuh Security Events - 151 Total Alerts, 17 Auth Failures

Alert Level Evolution Chart

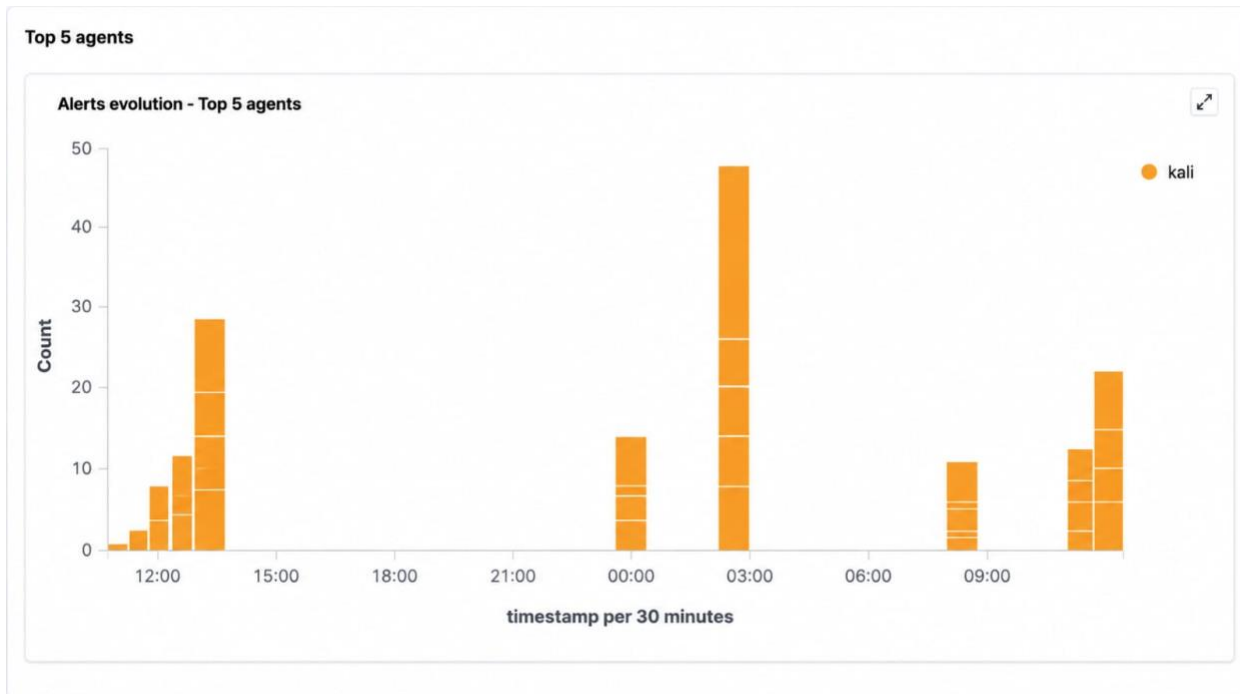
The alert chart showed different alert levels over time. Some periods had a higher number of alerts, indicating increased security activity that required attention.



Alert Level Evolution - Multi-Level Security Events Over 24 Hours

Top 5 Agents Alert Evolution

The Kali Linux agent generated most of the alerts. The chart showed when alert activity increased and helped identify busy monitoring periods.



Top 5 Agents - Alert Evolution Chart (Kali Agent)

Log Analysis

Kibana SSH Logs

Kibana Discover was used to search SSH logs. Several SSH-related events were found, including login failures and authentication activities.

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ sudo systemctl start ssh  
(kali@kali)-[~]  
$ sudo systemctl status ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)  
   Active: active (running) since Mon 2026-05-25 11:44:37 EDT; 2s ago  
  Invocation: 80e993f2f1454459ad0ae24ea8e3dbdd  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
  Process: 15654 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)  
 Main PID: 15656 (sshd)  
    Tasks: 1 (limit: 7312)  
  Memory: 2.1M (peak: 3.1M)  
     CPU: 60ms  
   CGroup: /system.slice/ssh.service  
           └─15656 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
May 25 11:44:37 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...  
May 25 11:44:37 kali sshd[15656]: Server listening on 0.0.0.0 port 22.  
May 25 11:44:37 kali sshd[15656]: Server listening on :: port 22.  
May 25 11:44:37 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

Kibana Discover - SSHD Logs, 17 Hits Identified

Failed Login Analysis

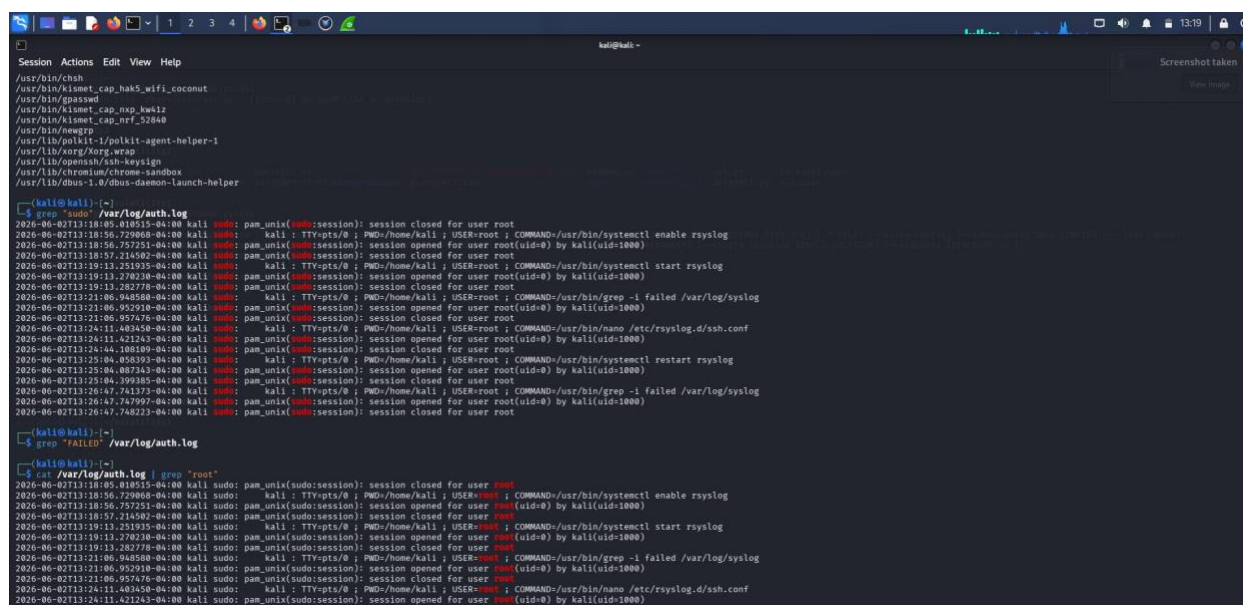
The auth.log file was checked using Linux commands. Failed login attempts from an invalid user account were detected, showing possible unauthorized access attempts.

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ping 127.0.0.1  
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:  
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=1.42 ms  
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.040 ms  
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.058 ms  
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.039 ms  
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.075 ms  
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.052 ms  
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.051 ms  
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.063 ms  
64 bytes from 127.0.0.1: icmp_seq=9 ttl=64 time=0.047 ms  
64 bytes from 127.0.0.1: icmp_seq=10 ttl=64 time=0.060 ms  
64 bytes from 127.0.0.1: icmp_seq=11 ttl=64 time=0.051 ms  
64 bytes from 127.0.0.1: icmp_seq=12 ttl=64 time=0.034 ms  
64 bytes from 127.0.0.1: icmp_seq=13 ttl=64 time=0.075 ms  
64 bytes from 127.0.0.1: icmp_seq=14 ttl=64 time=0.059 ms  
^C  
--- 127.0.0.1 ping statistics ---  
14 packets transmitted, 14 received, 0% packet loss, time 1327ms  
rtt min/avg/max/mdev = 0.034/0.152/1.416/0.350 ms  
  
(kali@kali)-[~]  
$ ping 127.0.0.1  
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:  
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.063 ms  
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.061 ms  
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.049 ms  
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.062 ms  
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.050 ms  
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.034 ms  
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.052 ms  
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.043 ms  
^C  
--- 127.0.0.1 ping statistics ---  
8 packets transmitted, 8 received, 0% packet loss, time 7157ms  
rtt min/avg/max/mdev = 0.034/0.052/0.063/0.009 ms  
  
(kali@kali)-[~]  
$ grep "Failed" /var/log/auth.log  
2026-06-02T13:19:25.891808-04:00 kali sshd-session[40772]: failed password for invalid user wronguser from 10.193.143.192 port 52630 ssh2  
2026-06-02T13:19:25.891855-04:00 kali sshd-session[40772]: failed password for invalid user wronguser from 10.193.143.192 port 52630 ssh2  
2026-06-02T13:19:35.159386-04:00 kali sshd-session[40772]: failed password for invalid user wronguser from 10.193.143.192 port 52630 ssh2  
  
(kali@kali)-[~]  
$ sudo grep "sudo" /var/log/auth.log  
[sudo] password for kali:  
2026-06-02T13:18:48.818115-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:18:56.729868-04:00 kali sudo:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/systemctl enable rsyslog  
2026-06-02T13:18:56.757251-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
2026-06-02T13:18:57.216492-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
```

Terminal - ping Test & grep Failed Logins in auth.log

Sudo Activity Analysis

The auth.log file also showed sudo activities. These logs recorded commands executed with administrator privileges and helped track user actions.

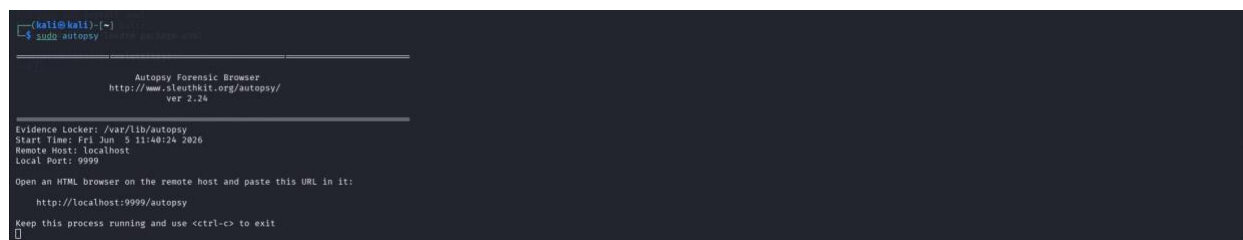


```
kali@kali: ~  
Session Actions Edit View Help  
/usr/bin/chsh  
/usr/bin/kismet_cap_hak5_wifi_cocmut  
/usr/bin/gpasswd  
/usr/bin/kismet_cap_nxp_kw412  
/usr/bin/kismet_cap_nrf_52848  
/usr/bin/newgrp  
/usr/lib/polkit-1/polkit-agent-helper-1  
/usr/lib/ksng/korg-wrap  
/usr/lib/openssh/ssh-keysign  
/usr/lib/chromium/chrome-sandbox  
/usr/lib/ibus-1.0/ibus-daemon-launch-helper  
  
kali@kali: ~  
$ grep "sudo" /var/log/auth.log  
2026-06-02T13:18:05.010515-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:18:56.729068-04:00 kali sudo: kali: TTYpts/0 : PWD=/home/kali : USER=root : COMMAND=/usr/bin/systemctl enable rsyslog  
2026-06-02T13:18:56.757251-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
2026-06-02T13:18:57.214502-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:19:13.251335-04:00 kali sudo: kali: TTYpts/0 : PWD=/home/kali : USER=root : COMMAND=/usr/bin/systemctl start rsyslog  
2026-06-02T13:19:13.270238-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
2026-06-02T13:19:13.282778-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:21:06.948588-04:00 kali sudo: kali: TTYpts/0 : PWD=/home/kali : USER=root : COMMAND=/usr/bin/grep -i failed /var/log/syslog  
2026-06-02T13:21:06.952910-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
2026-06-02T13:21:06.957476-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:21:06.957476-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:24:11.403458-04:00 kali sudo: kali: TTYpts/0 : PWD=/home/kali : USER=root : COMMAND=/usr/bin/nano /etc/rsyslog.d/ssh.conf  
2026-06-02T13:24:11.421243-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
2026-06-02T13:24:44.108189-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:25:04.905093-04:00 kali sudo: kali: TTYpts/0 : PWD=/home/kali : USER=root : COMMAND=/usr/bin/systemctl restart rsyslog  
2026-06-02T13:25:04.907343-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
2026-06-02T13:25:04.909185-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:26:47.743373-04:00 kali sudo: kali: TTYpts/0 : PWD=/home/kali : USER=root : COMMAND=/usr/bin/grep -i failed /var/log/syslog  
2026-06-02T13:26:47.747997-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
2026-06-02T13:26:47.748223-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
  
kali@kali: ~  
$ grep "FAILED" /var/log/auth.log  
  
kali@kali: ~  
$ cat /var/log/auth.log | grep "root"  
2026-06-02T13:18:05.010515-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:18:56.729068-04:00 kali sudo: kali: TTYpts/0 : PWD=/home/kali : USER=root : COMMAND=/usr/bin/systemctl enable rsyslog  
2026-06-02T13:18:56.757251-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
2026-06-02T13:18:57.214502-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:19:13.251335-04:00 kali sudo: kali: TTYpts/0 : PWD=/home/kali : USER=root : COMMAND=/usr/bin/systemctl start rsyslog  
2026-06-02T13:19:13.270238-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
2026-06-02T13:19:13.282778-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:21:06.948588-04:00 kali sudo: kali: TTYpts/0 : PWD=/home/kali : USER=root : COMMAND=/usr/bin/grep -i failed /var/log/syslog  
2026-06-02T13:21:06.952910-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
2026-06-02T13:21:06.957476-04:00 kali sudo: pam_unix(sudo:session): session closed for user root  
2026-06-02T13:24:11.403458-04:00 kali sudo: kali: TTYpts/0 : PWD=/home/kali : USER=root : COMMAND=/usr/bin/nano /etc/rsyslog.d/ssh.conf  
2026-06-02T13:24:11.421243-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
```

Terminal - sudo Activity and Root Session Logs

Autopsy Launch

Autopsy Forensic Browser was started for forensic investigation. Additional authentication logs and system activities were reviewed during the analysis.



```
kali@kali: ~  
$ sudo autopsy  
  
Autopsy Forensic Browser  
http://www.sleuthkit.org/autopsy/  
ver 2.24  
  
Evidence Locker: /var/lib/autopsy  
Start Time: Fri Jun 5 11:48:24 2026  
Remote Host: localhost  
Local Port: 9999  
  
Open an HTML browser on the remote host and paste this URL in it:  
http://localhost:9999/autopsy  
  
Keep this process running and use <ctrl-c> to exit  
█
```

Terminal - Autopsy Launch & Extended Auth Log

User and Privilege Analysis

Commands such as whoami, id, sudo -l, and find were used to gather user and privilege information. The system permissions and SUID files were examined for security analysis.

```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]:~  
# ansari  
kali  
[kali@kali]:~  
# id  
uid=1000(kali) gid=1000(kali) groups=1000(kali),4(adm),20(dialout),26(cdrom),25(floppy),27(sudo),29(audio),38(dip),44(video),46(plugdev),100(users),101(netdev),102(scanner),104(blueetooth),113(lpadmin),122(wreshark),123(kaboxer),124(vb  
usr)  
[kali@kali]:~  
# sudo -l  
[sudo] password for kali:  
Matching Defaults entries for kali on kali:  
env_reset, secure_paths=/usr/local/sbin:/usr/bin:/usr/sbin:/usr/bin:/sbin:/bin, use_pty  
  
User kali may run the following commands on kali:  
(ALL : ALL) ALL  
  
[kali@kali]:~  
# find / -perm -4000 2s/dev/null  
/usr/sbin/mount.cifs  
/usr/sbin/mount.nfs  
/usr/bin/pppd  
/usr/bin/kismet_cap_ti_cc_2540  
/usr/bin/phexec  
/usr/bin/nfs-3g  
/usr/bin/passwd  
/usr/bin/chfn  
/usr/bin/mount  
/usr/bin/kismet_cap_rz_killerbee  
/usr/bin/rsh-redone-eh  
/usr/bin/rsh-redone-flogin  
/usr/bin/kismet_cap_nrf_mousejack  
/usr/bin/sudo  
/usr/bin/kismet_cap_nrf_51022  
/usr/bin/kismet_cap_ubertooth_one  
/usr/bin/kismet_cap_linux_bluetooth  
/usr/bin/kismet_cap_ti_cc_2531  
/usr/bin/wc  
/usr/bin/fusemount3  
/usr/bin/mount  
/usr/bin/kismet_cap_linux_wifi  
/usr/bin/chsh  
/usr/bin/kismet_cap_hak5_wifi_cocomut  
/usr/bin/gpasswd  
/usr/bin/kismet_cap_nxp_kw412  
/usr/bin/kismet_cap_nrf_52848  
/usr/bin/newgrp  
/usr/lib/polkit-1/polkit-agent-helper-1  
/usr/lib/openssh/ssh-keysign
```

Terminal - whoami, id, sudo -l, SUID Binary Enumeration

Root Session Logs

Root-level activities and scheduled CRON jobs were identified from system logs. Network connectivity tests were also performed to verify system communication.

[illegible]

```
kali@kali: ~$ ping 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.142 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.040 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.058 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.039 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.075 ms
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.052 ms
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.051 ms
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.093 ms
64 bytes from 127.0.0.1: icmp_seq=9 ttl=64 time=0.067 ms
64 bytes from 127.0.0.1: icmp_seq=10 ttl=64 time=0.046 ms
64 bytes from 127.0.0.1: icmp_seq=11 ttl=64 time=0.051 ms
64 bytes from 127.0.0.1: icmp_seq=12 ttl=64 time=0.034 ms
64 bytes from 127.0.0.1: icmp_seq=13 ttl=64 time=0.075 ms
64 bytes from 127.0.0.1: icmp_seq=14 ttl=64 time=0.059 ms
^C
127.0.0.1 ping statistics:
14 packets transmitted, 14 received, 0% packet loss, time 13277ms
rtt min/avg/max/mdev = 0.034/0.152/1.416/0.350 ms

kali@kali: ~$ ping 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.063 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.061 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.049 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.062 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.050 ms
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.034 ms
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.052 ms
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.043 ms
^C
127.0.0.1 ping statistics:
8 packets transmitted, 8 received, 0% packet loss, time 7157ms
rtt min/avg/max/mdev = 0.034/0.052/0.063/0.009 ms

kali@kali: ~$ grep "Failed" /var/log/auth.log
2020-06-02T13:19:25.891080-04:00 kali sshd-session[40772]: Failed password for invalid user wronguser from 10.191.143.192 port 52630 ssh2
2020-06-02T13:19:31.149355-04:00 kali sshd-session[40772]: Failed password for invalid user wronguser from 10.191.143.192 port 52630 ssh2
2020-06-02T13:19:35.159386-04:00 kali sshd-session[40772]: Failed password for invalid user wronguser from 10.191.143.192 port 52630 ssh2

kali@kali: ~$ sudo grep "sudo" /var/log/auth.log
2020-06-02T13:18:05.010515-04:00 kali sudo: pam_unix(sudo:session): session opened for user root by kali(uid=1000)
2020-06-02T13:18:56.729068-04:00 kali sudo:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/systemctl enable rsyslog
2020-06-02T13:18:56.757251-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
2020-06-02T13:18:57.214502-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
```

Terminal - cat auth.log grep root & CRON Sessions

Network Packet Analysis (Wireshark)

Wireshark was used to capture and analyze network traffic. Different filters were applied to identify network activities and possible security issues.

DNS Filter Analysis

The DNS filter was used to check DNS traffic. No DNS query and response activity was found, indicating possible DNS or network connectivity issues.

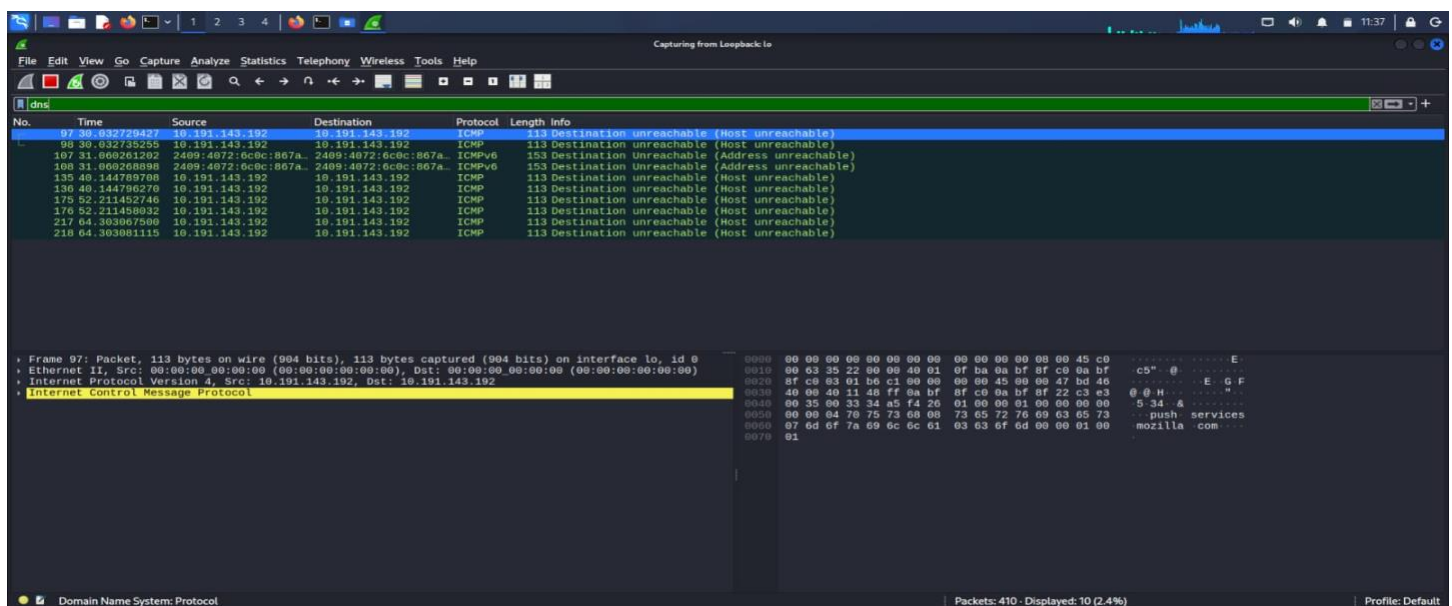
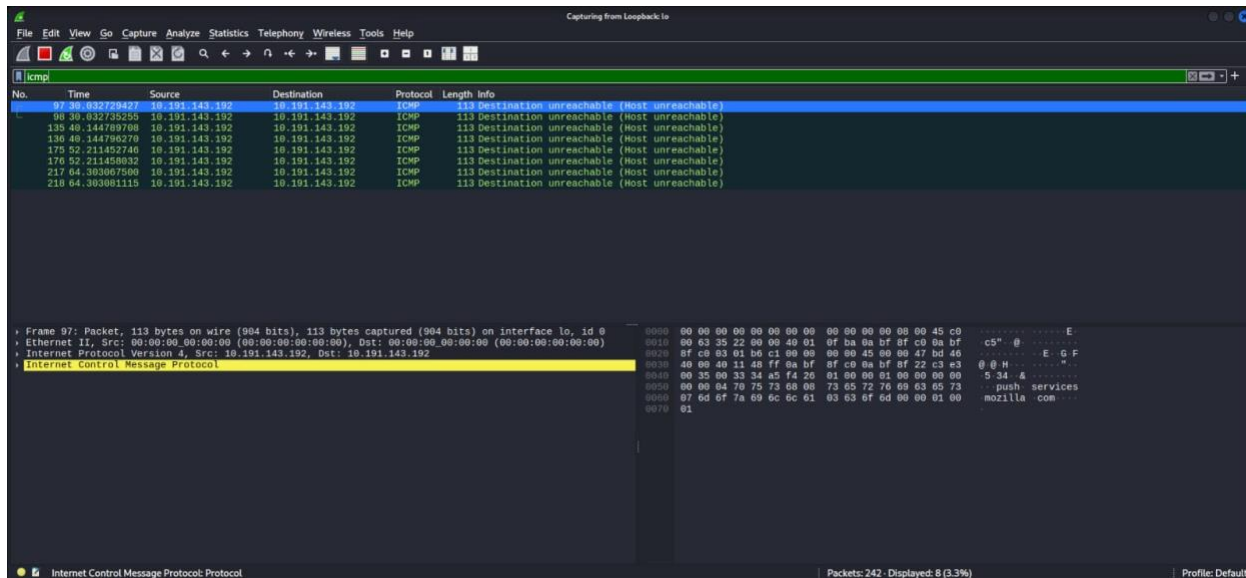


Figure 12: Wireshark - DNS Filter showing ICMP Unreachable Responses

ICMP Filter Analysis

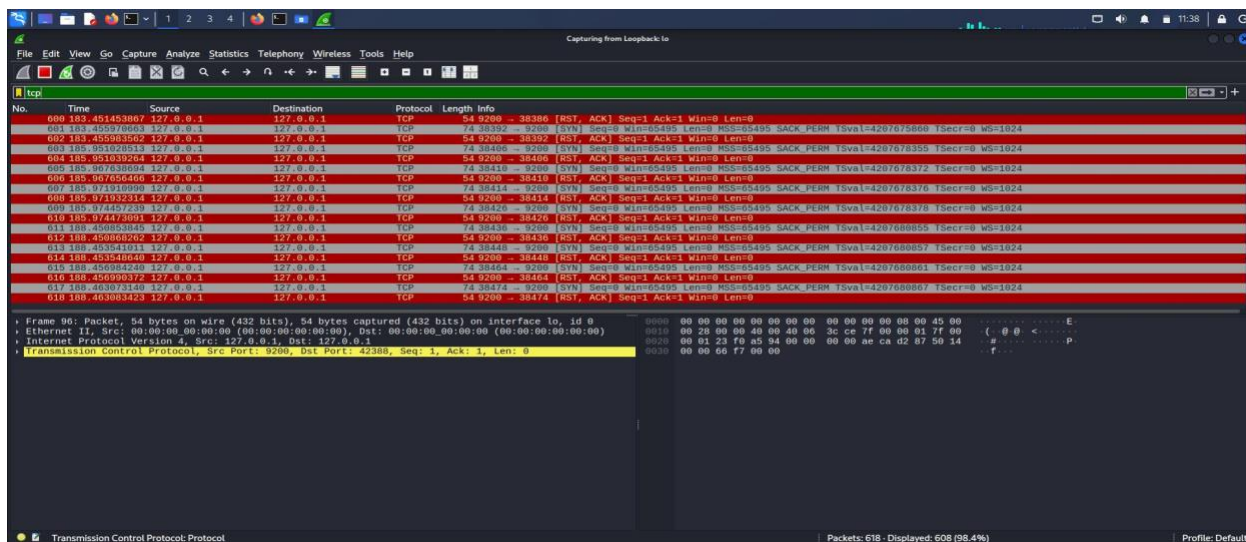
The ICMP filter showed several “Host Unreachable” packets. This indicated that the system was unable to reach certain network destinations.



Wireshark - ICMP Filter showing Host Unreachable Packets

TCP Filter Analysis

The TCP filter showed repeated connection attempts and responses. This pattern suggested possible port scanning or brute-force activity on the system.



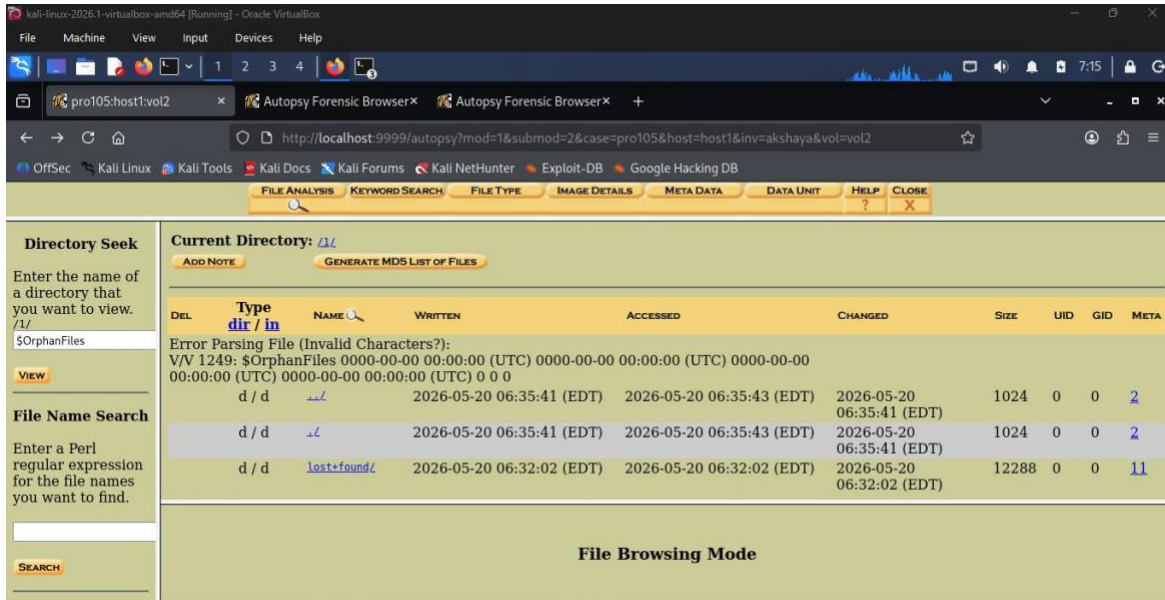
Wireshark - TCP Filter showing SYN/RST Brute Force Pattern

Digital Forensics (Autopsy)

Autopsy Forensic Browser was used to analyze disk images and examine files stored in the system.

File System Analysis

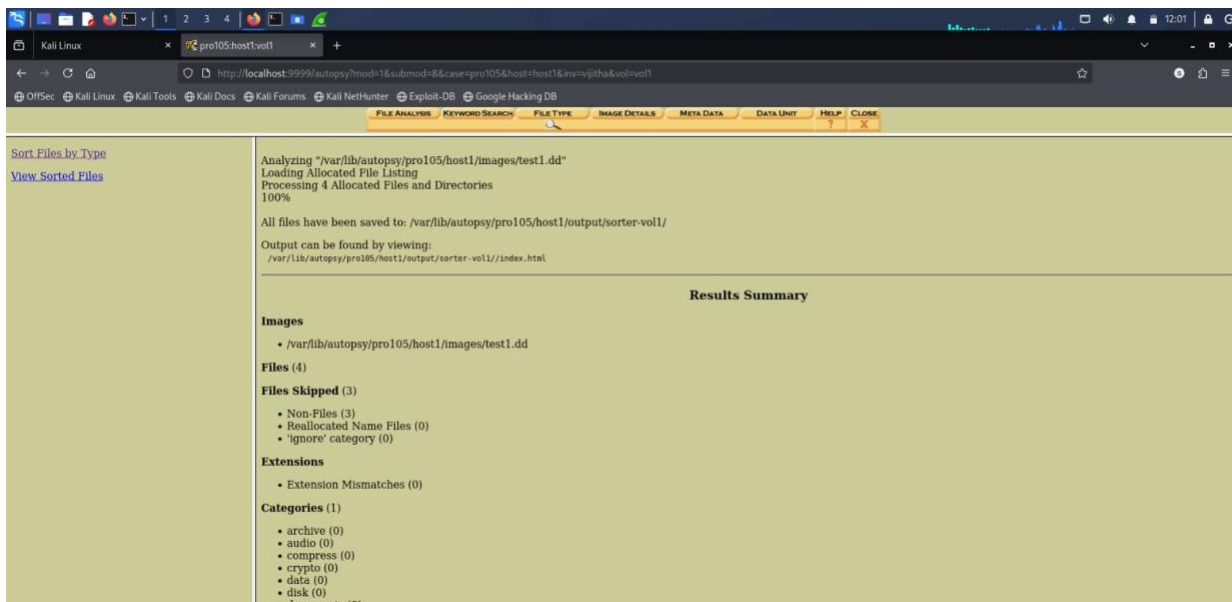
The disk image was explored to view folders and system directories. The file structure of the volume was successfully analyzed.



Autopsy - File System Directory Browser (vol2, /1/)

File Type Analysis

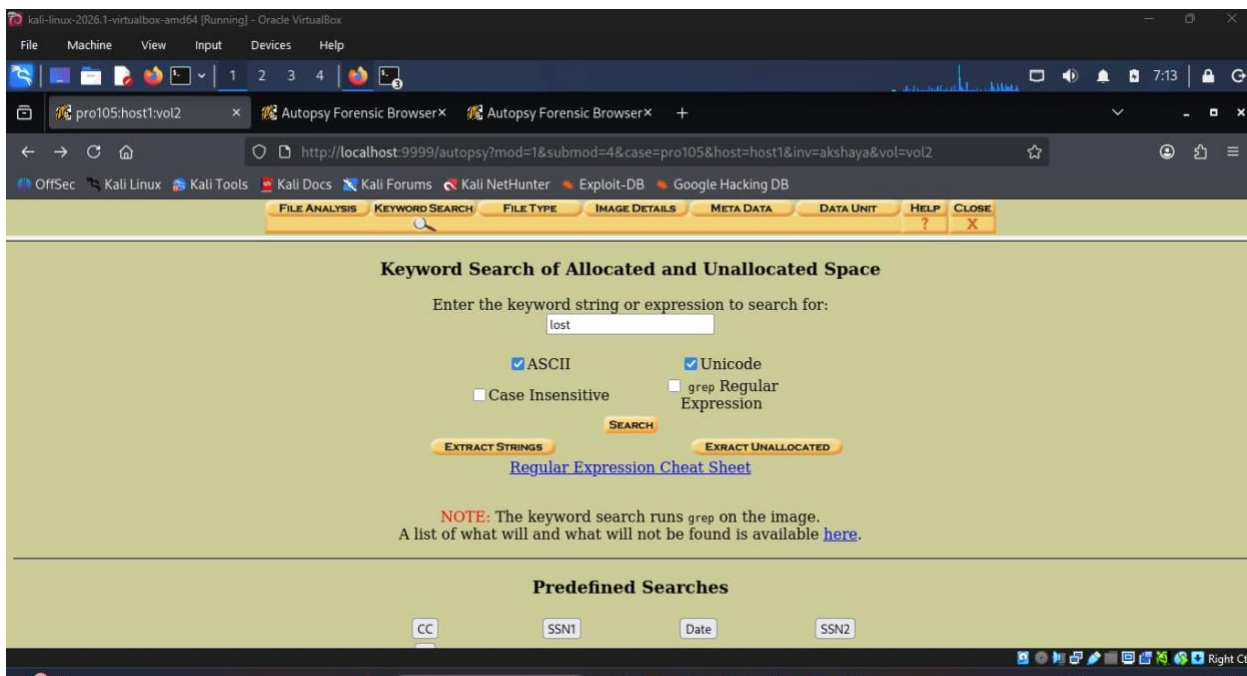
Autopsy categorized files based on their types. A text file was identified, and no suspicious file extension mismatches were found.



Autopsy – File Type Sort Results Summary (vol1)

Keyword Search

A keyword search was performed on the disk image to find relevant data. Both allocated and unallocated space were searched for possible evidence.



Autopsy - Keyword Search for 'lost' in vol2

Memory Forensics (Volatility Framework)

Volatility Framework was used to analyze the Windows memory dump file (memory.raw). Different plugins were used to examine running processes, network connections, and command-line activities.

Process Analysis

The process list showed several running Windows processes, including System, explorer.exe, services.exe, svchost.exe, and notepad.exe.

Network Connection Analysis

Active network connections were identified from the memory dump. Some processes were communicating with external IP addresses.

Command Line Analysis

The command-line investigation showed that notepad.exe had opened a file named "notes.txt" from the Desktop. This file was marked for further investigation.

```
C:\Users\user\Desktop>python vol.py -f memory.raw windows.pslist
Volatility Foundation Volatility Framework 2.6.1
```

Offset(V)	Name	Pid	PPid	Thds	Hnds	Sess	Wow64	Start	Exit
0x0000000003e52c060	System	4	0	106	556	N/A	False	2019-05-23 02:24:40 UTC+0000	-
0x0000000003e52d2d0	smss.exe	328	4	3	29	N/A	False	2019-05-23 02:24:40 UTC+0000	-
0x0000000003e531060	csrss.exe	456	448	10	377	0	False	2019-05-23 02:24:40 UTC+0000	-
0x0000000003e532060	wininit.exe	524	448	3	77	0	False	2019-05-23 02:24:40 UTC+0000	-
0x0000000003e53a060	csrss.exe	532	516	11	331	1	False	2019-05-23 02:24:40 UTC+0000	-
0x0000000003e6b5060	services.exe	612	524	8	219	0	False	2019-05-23 02:24:41 UTC+0000	-
0x0000000003e6b7060	lsass.exe	632	524	7	653	0	False	2019-05-23 02:24:41 UTC+0000	-
0x0000000003e71c060	svchost.exe	740	612	16	1,038	0	False	2019-05-23 02:24:41 UTC+0000	-
0x0000000003e71e060	svchost.exe	852	612	22	1,191	0	False	2019-05-23 02:24:41 UTC+0000	-
0x0000000003e72a060	winlogon.exe	868	524	4	127	1	False	2019-05-23 02:24:41 UTC+0000	-
0x0000000003e734060	explorer.exe	1000	868	37	2,215	1	False	2019-05-23 02:24:41 UTC+0000	-
0x0000000003e9e3060	notepad.exe	3060	1000	2	35	1	False	2019-05-23 02:25:10 UTC+0000	-

```
C:\Users\user\Desktop>python vol.py -f memory.raw windows.netscan
Volatility Foundation Volatility Framework 2.6.1
```

Offset(V)	Proto	Local Address	Foreign Address	State	Pid	Owner	Created
0x0000000003f2b5b30	TCPv4	0.0.0.0:135	0.0.0.0:0	LISTENING	740	svchost.exe	-
0x0000000003f2b5f60	TCPv4	0.0.0.0:445	0.0.0.0:0	LISTENING	4	System	-
0x0000000003f2b68b0	TCPv4	0.0.0.0:3389	0.0.0.0:0	LISTENING	852	svchost.exe	-
0x0000000003f2b6d30	TCPv4	192.168.1.10:49158	20.190.2.146:443	ESTABLISHED	1000	explorer.exe	2019-05-23 02:31:15 UTC+0000
0x0000000003f2b70f0	TCPv4	192.168.1.10:49160	13.107.21.200:443	ESTABLISHED	1000	explorer.exe	2019-05-23 02:31:16 UTC+0000
0x0000000003f2b73f0	TCPv4	192.168.1.10:49162	52.114.132.31:443	ESTABLISHED	1000	explorer.exe	2019-05-23 02:31:18 UTC+0000
0x0000000003f2b7760	UDPv4	0.0.0.0:5353	*:*	-	740	svchost.exe	-
0x0000000003f2b7c40	UDPv4	192.168.1.10:137	*:*	-	4	System	-
0x0000000003f2b7e10	UDPv4	192.168.1.10:138	*:*	-	4	System	-

```
C:\Users\user\Desktop>python vol.py -f memory.raw windows.cmdline
Volatility Foundation Volatility Framework 2.6.1
```

Offset(V)	Pid	Process	CommandLine
0x0000000003e52d2d0	328	smss.exe	\SystemRoot\System32\smss.exe
0x0000000003e531060	456	csrss.exe	%SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows
0x0000000003e532060	524	wininit.exe	wininit.exe
0x0000000003e6b5060	612	services.exe	C:\Windows\system32\services.exe
0x0000000003e6b7060	632	lsass.exe	C:\Windows\system32\lsass.exe
0x0000000003e734060	1000	explorer.exe	C:\Windows\Explorer.EXE
0x0000000003e9e3060	3060	notepad.exe	notepad.exe C:\Users\user\Desktop\notes.txt

```
C:\Users\user\Desktop>_
```

Key Findings & Incident Summary

SSH Brute Force Attack

- Failed SSH login attempts were detected.
- Wazuh generated authentication failure alerts.
- The activity showed signs of a brute-force attack.

Privilege Escalation Risk

- The Kali user had sudo access.
- SUID files were found on the system.
- Admin activities were recorded in the logs.

Network Issues

- Some network connections were unreachable.
- DNS was not working properly.
- Network scanning activity was observed.

Memory Forensics Findings

- Running processes were identified.
- Active network connections were found.
- System activities were reviewed from memory data.

Disk Forensics Findings

- A text file was recovered from the disk image.
- System directories were analyzed.
- No suspicious file extensions were found.

Recommendations

- Block suspicious IP addresses.
- Enable SSH protection tools such as Fail2Ban.

- Restrict unnecessary sudo access.
- Enable Multi-Factor Authentication (MFA).
- Monitor network traffic regularly.
- Configure SIEM alerts for critical events.
- Perform regular forensic backups and log reviews.
- Preserve evidence for future investigations.

Conclusion

This capstone project demonstrated a complete SOC and digital forensics investigation process. Wazuh was used for security monitoring, Wireshark for network analysis, Autopsy for disk forensics, and Volatility for memory analysis. The project helped identify suspicious activities, analyze evidence, and generate an incident investigation report. It provided practical experience in security monitoring, incident response, and digital forensic analysis.